Andrew Liao

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Education

New York University — New York, NY
BA Computer Science & Data Science — *Sep 2021 - Dec 2025*

Carnegie Mellon University — Pittsburgh, PA
MS AI Engineering (CompBME) - Thesis Track — *Aug 2026 - May 2028*

Publications

- **Y. Liao**, J. Gonzalez, Z. S. Chen. *Characterizing Breathing–Whisking Rhythms and Decoding Whisker Motion Energy.* Abstract accepted for poster presentation, **Neuroscience 2025**.
- Daniel Doan, **Andrew YS Liao**, Arnav Pallem, Kevin Zhu, Sunishchal Dev, Ashwinee Panda, Shreyas Sunil Kulkarni. *Emergent Misalignment in Mixture-of-Experts Models.* **Socially Responsible and Trustworthy Foundation Models, NeurIPS 2025.**
- **Ying-Shao Liao**, Hsin-Yun Hsu. *One-step photochemical crosslinking of native proteins is feasible in tyrosine-rich bovine serum albumin.* Journal of Emerging Investigators, Vol. 4 (May 7, 2021).

Research

New York University — GRAIL Lab — New York, NY
Undergraduate Researcher — *Jan 2025 – Jul 2025*

- **Robot Utility Models (RUMs) — Manipulation Policy for Zero-Shot Deployment**
  * Owned end-to-end development of **place** and **drop** manipulation skills on a **Hello Robot** mobile manipulator: 3D-printed a custom gripper tool, established data collection protocols, and built cleaning pipelines to curate **1,000+ demonstration videos** across diverse object shapes, colors, and sizes.
  * Trained visuomotor policies on **NYU's Greene HPC** (multi-GPU), achieving robust zero-shot generalization — the robot reliably placed and dropped arbitrary graspable objects into a target zone (red-tape-marked box) without object-specific fine-tuning.
  * Iterated through full **data collection** → **cleaning** → **training** → **deployment** → **re-collection** cycles; diagnosed sim-to-real failure modes and resolved distribution shifts via targeted data augmentation and retraining.

NYU Langone Health — CN^3 Lab — New York, NY
Undergraduate Researcher — *Feb 2025 – Present*

- **Multi-Camera Behavioral Video Pipeline & Neural Decoding** — Advisors: Joaquin Gonzalez, Zhe Sage Chen
  * Designed a **memory-optimized video processing pipeline** synchronizing **80+ single/dual-camera recordings** across 12 labs (100+ animals); built aligned feature trajectories for nose PCA, whisker energy, pupil tracking, and tongue detection — directly analogous to multi-sensor fusion in embodied systems.
  * Implemented **memmap-based lag construction** and memory-bounded ridge solvers (chunked $X^\top X$ / $X^\top y$ accumulation), reducing per-session RAM from **>50 GB to 10 GB** and achieving **12–30× speedups**; complexity reduced from $O(nd)$ to $O(d^2)$ for 4,500-feature models.
  * Produced brain-wide **R^2 decoding maps** (peak $R^2 \approx 0.49$) from 240k-sample sessions; first-author abstract accepted at **Neuroscience 2025**. This project has been accepted at the **International Brain Laboratory** for further mentorship and development

Algoverse — New York, NY
Researcher (Advisors: Shreyas Kulkarni, Ryan Lagasse, Sunishchal Dev) — *Jul 2025 – Oct 2025*

- **Co-first author** of *Emergent Misalignment in Mixture-of-Experts Models* (Accepted at **AAAI AIGOV 2026, NeurIPS FM 2025**)
- Designed a **30-run fine-tuning grid** across Mixtral-8×7B, Qwen3-30B, and GPT-oss-20B using **4-bit QLoRA**; discovered a negative correlation between expert sparsity and emergent misalignment. Validated via LLM-as-a-judge (GPT-4o, DeepSeek-V3) and StrongREJECT benchmark.

New York University — New York, NY
Research Volunteer (Advisors: Parijat Dube, Rajarshi Das) — *Jan 2025 – Aug 2025*

- **Clone-Structured Cognitive Graphs (CSCG):** Implemented the **eFeX algorithm in JAX** for latent graph recovery in aliased gridworld environments. Achieved tractable learning on **50×50 grids with 4-state aliasing**, reducing recovery time from exponential (random-walk baseline) to near-linear complexity.

NYU Langone Health — SUDCRRC — New York, NY
Undergraduate Researcher — *Feb 2025 – Present*

- **SUDC Risk Modeling:** Led ML analysis of a 346-case clinical registry using **FAMD/MCA**, **UMAP + K-Means** clustering (bootstrap ARI ≈ 0.95), and **CatBoost** ensembles (AUC = 0.78). First-author manuscript in preparation.

Technical Projects

CartPole Stabilization via Deep Reinforcement Learning — *Dec 2024*

- Built a full RL stack for continuous control: implemented a **custom CartPole environment**, designed deep neural network policies, and implemented **REINFORCE** with baseline and **Actor-Critic (GAE)** from scratch in PyTorch.
- Engineered a novel state representation decoupling acceleration dependencies while preserving the Markov property; achieved **475/500 average return** in just **120 training episodes**. Directly applicable to drone stabilization and control policy learning.